INCH-POUND

MIL-DTL-32630
3 July 2019

DETAIL SPECIFICATION

CONDUCTOR, ELECTRICAL, STRANDED, UNINSULATED CARBON-BASED CONDUCTIVE FIBER, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stranded electrical conductors fabricated from four types of uninsulated carbon-based conductive fiber for use in aerospace and other applications. Carbon-based conductive fibers are suitable for use in uninsulated and insulated conductors. Non-conductive fibers that are metalized are also considered to be carbon-based conductors in this specification.

1.2 Classification. Conductors are classified by the following types of fiber material they contain (see 3.1 and 6.2).

1.2.1 Types. Conductors are of the following types and compositions:

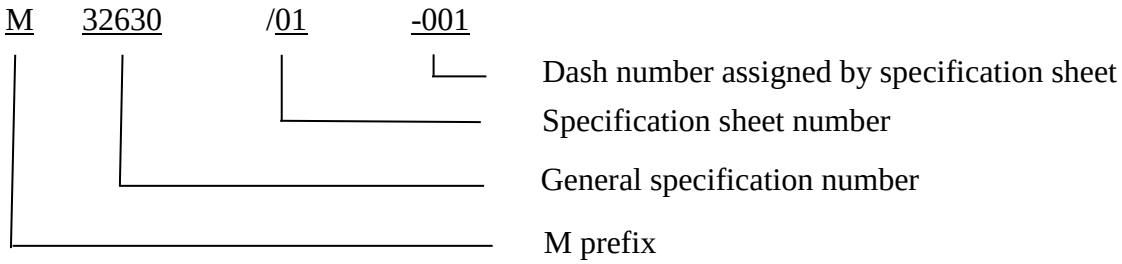
Type I - Conductors composed of metal coated aramid/para-aramid fiber (MCAF) (see 3.3.1).

Type II - Conductors composed of metal coated poly (p-phenylene-2, 6-benzobisoxazole (PBO) fiber (see 3.3.2).

Type III - Conductors composed of metal coated liquid crystal polymer (LCP) fiber (see 3.3.3).

Type IV - Conductors composed of carbon nanotube (CNT) fiber (see 3.3.4).

1.3 Part or Identifying Number (PIN). The PIN consists of the letter “M” followed by the specification number, the associated slash sheet number and a sequentially assigned dash number:



Comments, suggestions, or questions on this document should be addressed to: Naval Air Systems Command (Commander, Naval Air Warfare Center Aircraft Division, Code 4.1.2.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100) or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32630/1	-	Conductor, Electrical, Stranded, Uninsulated, Carbon Based, Conductive Fiber, 0.014 Nominal Diameter, Inclusive, -65 ºC to +150 ºC
MIL-DTL-32630/2	-	Conductor, Electrical, Stranded, Uninsulated, Carbon Based, Conductive Fiber, 0.010 Nominal Diameter, Exclusive, -65 ºC to +150 ºC
MIL-DTL-32630/3	-	Conductor, Electrical, Stranded, Uninsulated, Carbon Based, Conductive Fiber, 0.010 Nominal Diameter, -200 ºC to +200 ºC

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-202-107	-	Method 107, Thermal Shock
MIL-STD-202-208	-	Method 208, Solderability
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES (IPC)

IPC J-STD-002	Solderability Tests for Component Leads, Terminations, Lugs, Terminals and Wires

(Copies of this document are available online at http:/www.ipc.org.)

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ASTM INTERNATIONAL

ASTM B117	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B470	Standard Specification for Bonded Copper Conductors for Use in Hookup Wires for Electronic Equipment
ASTM D885/D885M	Standard Test Methods for Tire Cords, Tire Cords Fabrics, and Industrial Filament Yarns Made from Manufactured Organic-Base Fibers
ASTM E595	Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment
ASTM E662	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials

(Copies of these documents are available online at www.astm.org.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES INTERNATIONAL

NCSL-Z540.3	Requirements for the Calibration of Measuring and Test Equipment.

(Copies of this document are available online at http://ncsli.org.)

SAE INTERNATIONAL

SAE AS4373	Test Methods for Insulated Electric Wire

(Copies of this document are available online at http:/www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Conductors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.5 and 6.3).

3.3 Conductor material. The conductor base material (core) shall be one of the types specified herein (see 1.2). Multiple type conductors are covered by a single specification sheet provided that each conductor meets the requirements specified on the specification sheet. Prior approval to use a substitute material must be obtained from the qualifying activity. When a definite material is not specified, a material shall be used that will enable the finished products to meet the requirements of

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this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Type I conductor composed of metal coated para-aramid fibers. Each type I conductor shall be composed of a bundle of individual metal coated aramid fibers (MCAF) used as a shield or as a primary conductor.

3.3.2 Type II conductor composed of metal coated poly (p-phenylene-2, 6-benzobisoxazole (PBO) fibers. Each type II conductor shall be composed of a bundle of individually metal coated PBO Fibers (MCPF) as a shield or as a primary conductor.

3.3.3 Type III conductor composed of metal coated liquid crystal polymer (LCP) fibers. Each type III conductor shall be composed of a bundle of individually metal coated LCP Fibers (MCLF) as a shield or as primary conductor.

3.3.4 Type IV conductor composed of carbon nanotube (CNT) fibers. Each type IV conductor shall be composed of CNT fibers. CNT fibers shall be made up of conductive multi-walled carbon nanotubes or single walled carbon nanotubes. CNT fiber does not require coating, but may be metalized as a shield or as a primary conductor.

3.4 Coating materials. Conductive coating materials shall be as specified in the specification sheet. Carbon-based conductive fiber may incorporate one or more conductive coatings.

3.4.1 Uncoated (N) fibers. Carbon-based conductive fibers that are inherently conductive may be supplied uncoated. Uncoated fibers shall be designated as N.

3.4.2 Silver-coated (SC) fibers. Silver-coated fibers shall have a coating thickness adequate to ensure 95 percent solderability (see 3.6.18), when tested in accordance with 4.7.18. Silver-coated fibers shall be designated as SC.

3.4.3 Nickel-coated fibers (NC). Nickel-coated fibers shall have a coating thickness adequate to ensure acceptable results for salt fog testing in accordance with 4.7.15 for the duration specified by the applicable specification sheet. Nickel-coated fibers shall be designated as NC.

3.4.4 Heavy nickel-coated fibers (NHC). Heavy nickel-coated fibers shall have a coating thickness adequate to ensure acceptable results for salt fog testing in accordance with 4.7.15 for the duration specified by the applicable specification sheet. Heavy nickel-coated fibers shall be designated as NHC.

3.5 Design and construction. Unless otherwise specified (see 3.1), conductors shall be of the design and construction specified herein.

3.5.1 Length. There shall be no more than one single end splice in any 600 feet length of conductor. The splice shall retain 90 percent of the fiber’s strength. No spool of material shall contain more than two splices.

3.5.2 Twists per inch. The twisting and the number of twists per inch of the conductor shall be as required by the applicable specification sheet.

3.5.3 Multi-fiber conductors. Large conductors shall be constructed using balanced ply or bunch stranded constructions. No metallic coatings are permitted over the plied or stranded conductor after plying or stranding.

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3.5.3.1 Balance ply. The balance ply shall be as specified in the specification sheet, if a specific construction is required.

3.5.4 Workmanship. The conductor shall be manufactured and processed in such a manner as to be visually uniform in quality, and free of multiple broken filaments, significant oxidation, uneven color, and other defects that will affect life and reliability.

3.6 Performance requirements.

3.6.1 Diameter. The diameter of the conductor shall be as specified by the applicable specification sheet. The diameter shall be measured in accordance with 4.7.2.

3.6.2 Maximum DC resistance. The maximum DC resistance of the conductor shall be as specified by the applicable specification sheet. The DC resistance shall be measured in accordance with 4.7.3.

3.6.3 Fusing time. The fusing time of the conductor shall be measured in accordance with 4.7.4 and shall meet the applicable specification sheet requirements.

3.6.4 Break load. The minimum average break load of the conductor shall be as specified by the applicable specification sheet. The break load shall be measured in accordance with 4.7.5.

3.6.5 Elongation. The minimum elongation of the conductor shall be as specified by the applicable specification sheet. The elongation shall be measured in accordance with 4.7.5.

3.6.6 Flammability. When tested in accordance with 4.7.6, the conductor shall be self-extinguishing within 3 seconds after removal of the flame. Flame travel shall be less than 3 inches, and there shall be no flaming of tissue.

3.6.7 Flex life. When tested in accordance with 4.7.7, the conductor shall exhibit an average flex life specified in the specification sheet. If not specified, the conductor shall exceed greater than 500 cycles and not break or exceed the DC resistance specified in the applicable specification sheet. The test shall record the mandrel diameter, weight applied during the test, the deflection angle, the flex rate, and the number of cycles for each test specimen.

3.6.8 Outgassing. When tested in accordance with 4.7.8, the conductor total mass loss (TML) shall be less than 1 percent and less than 0.1percent of collected volatile condensable materials (CVCM).

3.6.9 Thermal shock. When tested in accordance with 4.7.9, the conductor shall not exceed the maximum DC resistance specified in the applicable specification sheet, and break strength shall not decrease by more than 30 percent of the initial value.

3.6.10 Life. When tested in accordance with 4.7.10, the conductor shall not rupture when pull tested and not exceed the DC resistance specified in the applicable specification sheet.

3.6.11 Smoke density. When the conductor is tested in accordance with 4.7.11, the Average Optical Density (D_s) of 10 shall not be exceeded after 20 minutes.

3.6.12 Smoke toxicity. When the conductor is tested in accordance with 4.7.12, the quantities of carbon monoxide (CO), nitric oxides, hydrogen cyanide (HCN), sulfur dioxide (SO_2), hydrogen chloride (HCl), and hydrogen fluoride (HF) released shall meet the table I requirements.

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TABLE I. Smoke toxicity requirements.

Material	Maximum acceptable amount released parts per million (ppm)
Carbon Monoxide (CO)	3500
Nitric Oxides (NO_x)	100
Hydrogen Cyanide (HCN)	150
Sulfur Dioxide (SO_2)	100
Hydrogen Chloride (HCl)	500
Hydrogen Fluoride (HF)	200

3.6.13 Fungus resistance. When tested in accordance with 4.7.13, the conductor shall exhibit no evidence of fungus growth and not exceed the maximum DC resistance specified in the applicable specification sheet.

3.6.14 Water absorption. When tested in accordance with 4.7.14, the conductor shall meet levels specified in the applicable specification sheet. If not specified, the conductor shall exhibit no more than 7 percent weight change.

3.6.15 Salt fog. When tested in accordance with 4.7.15, the conductor shall not exceed the maximum DC resistance specified in the specification sheet, and break strength shall not decrease by more than 25 percent of the initial value.

3.6.16 Fluid resistance. When tested in accordance with 4.7.16, the conductor after exposure shall not exceed the maximum DC resistance specified in the applicable specification sheet. The break strength and elongation shall not decrease by more than 25 percent of the initial value, and the weight and diameter shall not change by more than 5 percent of the initial value.

3.6.17 Termination crimp. When tested in accordance with 4.7.17, the conductor shall exceed the crimp termination break load of 5 pounds unconditioned and 4 pounds after life testing (see 4.7.10).

3.6.18 Solderability (SC fibers only). When tested in accordance 4.7.18, the conductor shall have a minimum solderability of 95 percent.

3.6.19 Coating workmanship. When examined in accordance with 4.7.19, the conductor shall exhibit no evidence of plating delamination.

3.6.20 Cross-sectional area. When measured in accordance with 4.7.20, the conductor shall meet the minimum cross-sectional area requirements in the applicable specification sheet.

3.6.21 Temperature coefficient of resistance. Temperature coefficient of resistance for each conductor material shall be determined in accordance with 4.7.21 to normalize resistance readings to 20 °C (68 °F). Temperature coefficient of resistance shall be as specified in the applicable specification sheet.

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4. VERIFICATION

4.1 Classification of inspections. The examinations and tests specified herein are classified as follows:

a. Qualification inspection (see 4.5).

b. Conformance inspections (see 4.6).

4.2 Test equipment and inspection facilities. The manufacturer shall establish and maintain testing and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspections. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with NCSL-Z540.3.

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202, General Requirements.

4.4 Cross-section inspection. Coatings shall be evaluated for uniformity and cross-sectional area using the following procedure.

4.4.1 Test specimen. The test specimen shall consist of a finished conductor in accordance with the applicable specification sheet. The test specimen shall be a size 26 conductor, unless otherwise specified in the applicable specification sheet.

4.4.2 Mounting. The test specimen cross-sections shall be mounted in a low-heat casting resin, encapsulating the sample for sectioning and polishing.

4.4.3 Polishing. The test specimen shall be ground and polished using the appropriate pads and paste to achieve a surface finish of 2 µinches or better.

4.4.4 Examination. Examine the finished cross-section of the conductor under at least 400X magnification. Plating thickness may be measured directly. Cross-sectional area of the conductor with a non-conductive core must be differentiated from the cross-sectional area of conductor coating on the core.

4.5 Qualification inspection. Qualification inspection shall consist of the examinations and tests in table II. The qualifying activity shall perform or witness the particular examinations or tests as noted in table II and the supplier shall perform the remaining tests. Unless otherwise specified, testing shall be conducted on a size 26 conductor.

4.5.1 Qualification inspection process. For guidance on the qualification inspection process, see 6.3 and 6.3.1.

4.5.2 Failures. One or more failures shall be cause for refusal to grant qualification.

4.5.3 Qualification inspection samples. Samples submitted for testing shall use the same materials, manufacturing procedures, and methods of inspection as would be used to provide the conductor to a purchaser. A representative 1,000 ft. (92.9 m) continuous length of a size 26 conductor shall be provided to the qualification activity for evaluation for each specification sheet type for which qualification is requested. An alternative length or size of the conductor may be provided, upon the approval of the qualifying activity. The number of samples provided may be reduced if there is sufficient similarity between the specification sheet samples, with the approval of the qualification activity.

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TABLE II. Qualification tests.

Examinations and tests	Requirement paragraph	Test paragraph
Visual examination	3.3, 3.4, and 3.5	4.7.1
Diameter	3.6.1	4.7.2
DC resistance 1/	3.6.2	4.7.3
Fusing time	3.6.3	4.7.4
Break load and elongation 1/	3.6.4 3.6.5	4.7.5
Flammability	3.6.6	4.7.6
Flex life1/	3.6.7	4.7.7
Outgassing	3.6.8	4.7.8
Thermal shock	3.6.9	4.7.9
Life	3.6.10	4.7.10
Smoke density	3.6.11	4.7.11
Smoke toxicity	3.6.12	4.7.12
Fungus resistance	3.6.13	4.7.13
Water absorption	3.6.14	4.7.14
Salt fog	3.6.15	4.7.15
Fluid resistance	3.6.16	4.7.16
Termination crimp	3.6.17	4.7.17
Solderability	3.6.18	4.7.18
Coating workmanship (coated conductors only)	3.6.19	4.7.19
Cross-sectional area	3.6.20	4.7.20
Temperature coefficient of resistance	3.6.21	4.7.21

1/Tests to be performed by qualifying activity.

4.5.4 Retention of qualification. Retention of qualification inspection will occur every 36 months after the initial qualification date. The qualifying activity will perform or witness the particular tests as specified in table III. The qualifying activity may shift the reporting date to accommodate testing schedules, but no later than 18 months from the retention of qualification due date.

TABLE III. Retention of qualification tests.

Examinations and tests	Requirement paragraph	Test paragraph
Visual examination	3.3, 3.4, and 3.5	4.7.1
Diameter	3.6.1	4.7.2
DC resistance1/	3.6.2	4.7.3
Break load and elongation1/	3.6.4 3.6.5	4.7.5
Termination crimp	3.6.17	4.7.17
Solderability	3.6.18	4.7.18
Coating workmanship (coated conductors only)	3.6.19	4.7.19
Cross-sectional area	3.6.20	4.7.20

1/ Test to be performed by the qualifying activity.

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4.6 Conformance inspection. Conformance inspection shall consist of the examinations and tests in table IV. As a minimum, conformance shall be performed on every manufacturing lot (see 6.4.3) of conductors purchased in accordance with this specification. The absence of any inspection requirements as specified herein shall not relieve the supplier of the responsibility of assuring that all products comply with all specification requirements. Sampling in conformance inspection does not authorize submission of known defective material, either indicated or actual.

TABLE IV. Conformance tests.

Examinations and tests	Requirement paragraph	Test paragraph
Visual examination	3.3, 3.4, and 3.5	4.7.1
Diameter	3.6.1	4.7.2
DC resistance	3.6.2	4.7.3
Break strength and elongation	3.6.4 3.6.5	4.7.5
Solderability (silver-plated conductors only)	3.6.18	4.7.18

4.6.1 Conformance testing. The conformance tests in table IV shall be conducted for each manufacturing lot (see 6.4.3). The results of the tests will be evaluated by the qualifying activity and the conductors will be accepted if all tests are passed.

4.6.2 Test sample configuration. The test sample shall be a twisted size 26 conductor. In instances where the mechanical stability of twisted fiber is not sufficient resulting in excessive measurement uncertainty, braided or plied constructions of the fiber may be tested in lieu of twisted fiber.

4.7 Test methods.

4.7.1 Visual examination (see 3.3, 3.4, and 3.5). The conductor and fiber shall be visually inspected to determine conformance to the conductor material, conductor coating material, and workmanship requirements.

4.7.2 Diameter (see 3.6.1). The diameter of the conductor shall be measured at three locations along the length of each sample while the conductor is under 0.5-pound tension. Each measurement shall be the average of two readings taken 90 degrees apart on the conductor. Measurement devices capable of reading to the nearest 0.0001 inch shall be used.

4.7.3 DC resistance (see 3.6.2). The DC resistance of the conductor shall be measured in accordance with SAE AS4373, Test Method 403 at 20 °C ±1 °C (temperature correction is based on the measured temperature coefficient of resistance, 4.7.21) except that the results shall be reported in ohms/feet and ohms/m for all conductor sizes. The average DC resistance shall also be measured and recorded on three uninsulated size 26 conductors and any other wire sizes specified in the applicable specification sheet.

4.7.4 Fusing time (see 3.6.3). The fusing time of the conductor shall be measured in accordance with SAE AS4373, Test Method 511 and conducted on three samples of uninsulated size 26 conductor using the construction and current given in the specification sheet.

4.7.5 Break load and elongation (see 3.6.4 and 3.6.5). Break load and elongation tests of the conductor shall be conducted in accordance with ASTM D885/D885M, using a constant-rate-of-specimen extension method, except that the grip separation speed shall be approximately 2 inches per

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minute. The break load (lb.) and elongation shall be determined by the means of a recording chart, or other means, on the testing machine. Tests shall be performed on the whole conductor and the break load and elongation measured when the first fibers of the conductor break.

4.7.6 Flammability (see 3.6.6). Flammability tests shall be conducted in accordance with SAE AS4373, Test Method 801, except the test shall be conducted on three samples of uninsulated size 26 conductor at least 3 feet (0.91 meter) in length.

4.7.7 Flex life (see 3.6.7). The flex life of an uninsulated size 26 conductor shall be determined using procedures outlined in ASTM B470 flex life test method. The following modifications shall be made to the test. Radius shall be 6X based on a 0.180 inch (4.6 mm) diameter conductor. Weight applied during flex test shall be 2 pounds (0.91 Kg). The resistance shall be measured using a four-point probe setup while applying a weight of 2 pounds (0.91 Kg) and installed on to the flex tester. Testing shall be terminated when the specified number of cycles is exceeded, the sample breaks, or the DC resistance exceeds the initial value by more than 15 percent. A minimum of six specimens shall be tested and the average value and standard deviation determined.

4.7.8 Outgassing (see 3.6.8). Outgassing testing shall be conducted in accordance with ASTM E595 on three samples approximately 10 feet (3.05 m) in length of uninsulated coiled size 26 conductor.

4.7.9 Thermal shock (see 3.6.9). Thermal shock test shall be conducted in air in accordance with MIL-STD-202-107, test condition selected (A through F) and shall be based on the high temperature specified by the applicable specification sheet. Three samples each of approximately a 3-foot (0.91 m) length of uninsulated size 26 conductor shall be placed on PTFE or PTFE-coated mandrels that are 6 times the diameter of the uninsulated conductor with the ends secured and subjected to the test. Record DC resistance before and after the test and measure break strength.

4.7.10 Life test (see 3.6.10). Three samples approximately 3 feet (0.91 m) in length of uninsulated size 26 conductor shall be placed on PTFE or PTFE-coated mandrels. The mandrels shall be 6 times the diameter of the uninsulated conductor with the ends secured and then exposed to its maximum temperature specified by the applicable specification sheet for 1,000 hours followed by 10 cycles of the thermal shock test over the range specified in 4.7.9 and then followed by an 80 percent of break strength pull test applied for 30 seconds. The DC resistance shall be measured before and after the life test.

4.7.11 Smoke density (see 3.6.11). Smoke density testing shall be conducted in accordance with ASTM E662. Three samples of size 26 uninsulated conductors, each approximately 10 feet (3.05m) in length, with a minimum coil diameter of 1 ft. (0.305 m) shall be used for this test. The coiled conductor shall be hung in the chamber.

4.7.12 Smoke toxicity (see 3.6.12). During smoke density testing (see 4.7.11) or in a separate test, capture and measure the evolving gasses and check for the quantities of the following gases: carbon monoxide, nitrogen oxides, hydrogen cyanide, sulfur dioxide, hydrogen chloride, and hydrogen fluoride. Results shall be reported as part of the smoke density test.

4.7.13 Fungus resistance (see 3.6.13). The manufacturer may verify that all external materials are fungus resistant, by certification, if approved by the qualifying activity. If verification by certification is not approved by the qualifying activity, the manufacturer shall test conductors as follows: Three samples of size 26 uninsulated conductors, each approximately 3 feet (0.91 m) in length shall be placed on PTFE or PTFE-coated mandrels that are 6 times the diameter of the uninsulated conductor with the

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ends secured. Attach lead wires to each end of the test sample. Measure lead wire resistance prior to attaching to test sample. Measure the DC resistance (see 4.7.3) through the lead wires. Test the samples for fungus resistance in accordance with MIL-STD-810, Test Method 508. The test shall last a minimum of 28 days. After the exposure, inspect the samples for evidence of fungus growth and measure the DC resistance (see 4.7.3) through the lead wires after the test.

4.7.14 Water absorption (3.6.14). Three samples of size 26 uninsulated conductor each approximately 10 feet. (3.05m) in length shall be evaluated. Measure mass of the test sample. Immerse the coiled test sample in deionized water for 168 hours at 90 °C. Allow the sample to air dry for 24 hours. Measure mass of the test samples.

4.7.15 Salt fog (see 3.6.15). Three samples of size 26 uninsulated conductor each approximately 3 feet (0.91 m) in length shall be placed on PTFE or PTFE-coated mandrels that are 6 times the diameter of the wire with the ends secured and subjected to the test. Attach lead wires to each end of the test sample. Measure the DC resistance (see 4.7.3) through the lead wires. Measure DC resistance (see 4.7.3) of the sample. Hang three samples in a chamber and expose them to the salt fog environment in accordance with ASTM B117, for the duration specified by the applicable specification sheet. If the duration is not specified in the specification sheet, 500 hours shall be used. Clean samples as specified in ASTM B117 using distilled water and measure DC resistance (see 4.7.3) of the sample.

4.7.16 Fluid resistance (see 3.6.16). Fluid resistance testing shall be conducted in accordance with SAE AS4373, Test Method 601 on three uninsulated size 26 conductors approximately 24 inches (610 mm) long using the specified fluids. The bend test and voltage withstand tests shall not be conducted. The conductors shall be evaluated after exposure to all fluids listed in SAE AS4373, Test Method 601. The DC resistance of the conductor shall be measured before and after the fluid exposure to determine compliance with 3.6.16. The initial break strength and elongation of the unexposed conductor shall be measured and the percent change in break strength, elongation, weight, and diameter shall be calculated to determine compliance with 3.6.16.

4.7.17 Termination crimp (see 3.6.17). Three size 26 uninsulated conductors each approximately 3 feet (0.91 m) in length shall be tested. When terminated with an appropriately sized crimp termination and standard tool, the terminated conductor shall withstand life testing in accordance with 4.7.10.

4.7.18 Solderability (see 3.6.18). Solderability testing shall be conducted on three uninsulated conductors in accordance with MIL-STD-202-208, except wrapping the conductor in accordance with IPC J-STD-002, Test C is not necessary and steam aging is not required. No solderability testing is required for nickel-coated conductors.

4.7.19 Coating workmanship (coated conductors only) (see 3.6.19). A metallurgical cross-section of coated conductors shall demonstrate that the individual fibers are coated 360 degrees with metal. This examination shall be done using magnification of at least 400X.

4.7.20 Cross-sectional area (see 3.6.20). Examine a cross-section of conductor using at least 400X magnification and measure the cross-sectional area in at least three different locations to determine the cross-sectional area of the entire conductor.

4.7.21 Temperature coefficient of resistance (see 3.6.21). Temperature coefficient of resistance shall be given to correct for resistance when the conductor temperature is over 20 ˚C. The coefficient

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shall be reported as change in resistance/degree centigrade over the temperature range of the conductor above 20 ˚C. The change in resistance shall be determined using at least 4 measured resistance values over the conductor temperature range above 20 ˚C. More than one value shall be provided if the calculated rate of change gives an error greater than 5 percent of the measured resistance value.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or with the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files. CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Conductors covered by this specification are intended to be used for braided shielding and as electrical conductors in insulated wires. These conductors, when used in insulated wires are required to withstand extreme environments not present in commercial applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Type (see 1.2.1).

c. Part or Identifying Number (PIN) (see 1.3).

d. Quantity of conductor required (typical unit of measure for these materials is unit length).

e. Packaging requirements (see 5.1).

6.3 Qualification inspection. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Product List QPL No. 32630 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Naval Air Systems Command (Code 4.4.5.3), 48298 Shaw Road, Bldg. 1461, Patuxent River, MD 20670-1900 or QPL_Support@navy.mil. An online database of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification process. At the initiation of the qualification process, prior to testing, prospective suppliers must forward a written request for qualification to the qualifying activity (see 6.3). The qualifying activity will respond in writing with a formal "letter of authorization" providing detailed instructions for the submission of product samples. Testing cannot begin until the supplier has

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received this authorization letter. The qualification process is further described in the SD-6 “Provisions Governing Qualification.” Copies of this document are available online at https://assist.dla.mil.

6.4 Definitions.

6.4.1 Balanced ply conductors. A multi-fiber conductor constructed from single carbon-based conductive fibers where the twist (see 6.4.12) of the individual fibers is nullified by the twist used to combine the fibers together (see 6.5.2).

6.4.2 Bunch stranded conductor. A multi-fiber conductor that is constructed from single carbon based conductive fibers that are stranded together from any number of singles (or balance plied) fibers in a random pattern, twisted in one operation (see 6.5.3).

6.4.3 Manufacturing lot. A manufacturing lot consists of all conductors covered under one specification sheet, manufactured from the specified raw materials, processed under the same specification and procedures, produced by the same type of equipment, and submitted for inspection at one time (see 6.6).

6.4.4 Multi-fiber. A conductor constructed from multiple single fibers using a plied or bunch stranded construction. Braided rope constructions are also possible.

6.4.5 Sample. The sample is the group of sample units selected from the lot for purpose of inspection.

6.4.6 Sample unit. The sample unit consists of a single piece of finished conductor of sufficient length to permit all applicable examinations and tests.

6.4.7 Single carbon-based conductive fiber. A single carbon-based conductive fiber is a carbon-based conductive yarn comprising multiple filaments that may be described by a nominal “as-round” diameter, tow (number of filaments), and filament size.

6.4.8 Tow. A single carbon-based conductive fiber comprising multiple filaments. The number of filaments is the tow.

6.4.9 Unit of product. A unit is one continuous length of conductor.

6.5 Fiber and conductor design considerations.

6.5.1 As-round diameter. A single carbon-based conductive fiber may be described by an equivalent diameter when twisted providing a reference cross-sectional area. The as-round diameter is a convenient size reference and does not specify the cross-sectional area of the conductive material. Carbon-based conductive fibers of varying diameters may meet the same requirements. The range of diameters are specified in the individual specification sheet.

6.5.2 Balanced ply conductors. Balanced ply conductors use twisted singles with twist in either S or Z direction are combined with a twist in the Z or S direction (in effect, the direction opposite the direction of the twist of the singles as shown on figure 1).

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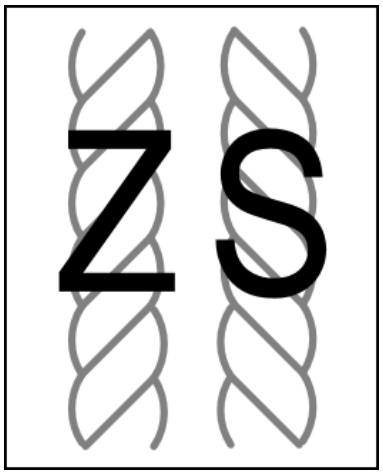


FIGURE 1. Z/S direction illustration.

Constructions of plied yarns may be repeated using S/Z/S or Z/S/Z constructions and so on. Plied conductors are referred to as shown by following examples: 2-ply = two single fibers combined to form a single conductor; 3/2-ply = three 2-ply fibers combined to form a single conductor. Carbon-based conductive fibers of varying balance ply constructions may meet the same requirements required balance ply (see 3.5.3.1 and 6.4.1).

6.5.3 Bunch stranded conductors. Bunch stranded conductors have all fibers twisted in the same direction with the same lay length resulting in a rougher surface and lower dimensional tolerance than balance plied constructions. Bunch stranded conductors are the least expensive type of stranding as it involves the least time and labor. This stranding method provides great flexibility. Carbon based conductive fibers of varying bunch strand constructions may meet the same requirements.

6.5.4 Bundle size. The size of the carbon based conductive fiber should be described for each wire size on the applicable specification sheet. The large number of very fine filaments together with the tendency of yarn bundles to flatten and spread, result in a cross section that is more rectangular than circular.

6.5.5 Coating thickness. Coating thickness of nickel and silver coating strands should be measured using optical or Scanning Electron Microscope (SEM) inspection of conductor cross-sections. Plating thickness may be measured directly. Cross-sectional area of coated carbon based conductive fiber with a non-conductive core must be differentiated from the cross-sectional area of conductor coating on the core.

6.5.6 Filament size. A single carbon based conductive fiber is comprised of multiple filaments. The filament size varies by base material type and may vary within any given base material type as well. Carbon based conductive fibers of varying filament sizes may meet the same requirements. The individual specification sheet should specify filament size, if a specific filament size is required.

6.5.7 Carbon-based fiber tow. Carbon-based conductive fibers of varying tows may meet the same requirements. The individual specification sheet specifies tow, if a specific tow is required.

6.5.8 Twist. Twisting fiber imparts strength to the conductor. It makes the fibers harder to pull apart lengthways and the conductor therefore harder to break. You can twist in either direction. If you

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hold a length of fiber at one end and twist the far end to the left, an “S” twist is created in the conductor. Twisting to the right creates a “Z” twist (see Figure 1).

6.6 Manufacturing lot. Each manufacturing lot of conductors should be a group identified by a common manufacturing record through all significant operations. Common manufacturing records and traceability should begin with machining and should include conductor assembly. Traceability of conductors to specific physical/chemical test reports of incoming raw material in not required (see 6.4.3).

6.7 Selection of sample units. Not more than one sample unit for each group of tests should be taken from a single unit of product.

6.8 Disposal of in-process waste. Caution must be taken during any plating, cleaning, descaling, passivation, or similar process.

6.9 Subject term (key word) listing.

Lightweight yarns
Nanotube

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

Project 6145-2018-050

Review activities:
Army - AR
Navy - SH
Air Force - 11, 20

NOTE: The activities above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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